

Question ID 7d6928bd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 7d6928bd

A cleaning service that cleans both offices and homes can clean at most **14** places per day. Which inequality represents this situation, where *f* is the number of offices and *h* is the number of homes?

- A. $f + h \leq 14$
- B. $f + h \geq 14$
- C. $f - h \leq 14$
- D. $f - h \geq 14$

ID: 7d6928bd Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that the cleaning service cleans both offices and homes, where *f* is the number of offices and *h* is the number of homes the cleaning service can clean per day. Therefore, the expression *f* + *h* represents the number of places the cleaning service can clean per day. It's also given that the cleaning service can clean at most **14** places per day. Since *f* + *h* represents the number of places the cleaning service can clean per day and the service can clean at most **14** places per day, it follows that the inequality $f + h \leq 14$ represents this situation.

Choice B is incorrect. This inequality represents a cleaning service that cleans at least **14** places per day.

Choice C is incorrect. This inequality represents a cleaning service that cleans at most **14** more offices than homes per day.

Choice D is incorrect. This inequality represents a cleaning service that cleans at least **14** more offices than homes per day.

Question Difficulty:
Easy

Question ID b64e2c7f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: b64e2c7f

Monarch butterflies can fly only with a body temperature of at least **55.0 degrees Fahrenheit (°F)**. If a monarch butterfly's body temperature is **51.3°F**, what is the minimum increase needed in its body temperature, in °F, so that it can fly?

- A. 1.3
- B. 3.7
- C. 5.0
- D. 6.3

ID: b64e2c7f Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that monarch butterflies can fly only with a body temperature of at least **55.0 degrees Fahrenheit (°F)**. Let x represent the minimum increase needed in the monarch butterfly's body temperature to fly. If the monarch butterfly's body temperature is **51.3°F**, the inequality $51.3 + x \geq 55.0$ represents this situation. Subtracting **51.3** from both sides of this inequality yields $x \geq 3.7$. Therefore, if the monarch butterfly's body temperature is **51.3°F**, the minimum increase needed in its body temperature, in °F, so that it can fly is **3.7**.

Choice A is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **53.7°F**, not **51.3°F**.

Choice C is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **50.0°F**, not **51.3°F**.

Choice D is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **48.7°F**, not **51.3°F**.

Question Difficulty:

Easy

Question ID b75f7812

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: b75f7812

Maria plans to rent a boat. The boat rental costs \$60 per hour, and she will also have to pay for a water safety course that costs \$10. Maria wants to spend no more than \$280 for the rental and the course. If the boat rental is available only for a whole number of hours, what is the maximum number of hours for which Maria can rent the boat?

ID: b75f7812 Answer

Rationale

The correct answer is 4. The equation $60h + 10 \leq 280$, where h is the number of hours the boat has been rented, can be written to represent the situation. Subtracting 10 from both sides and then dividing by 60 yields $h \leq 4.5$. Since the boat can be rented only for whole numbers of hours, the maximum number of hours for which Maria can rent the boat is 4.

Question Difficulty:
Easy

Question ID 208626df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 208626df

$$2\ell + 2w \leq 27$$

A rectangle has length ℓ and width w . The inequality gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to **27**. Which statement is the best interpretation of $(\ell, w) = (8, 3)$ in this context?

- A. If the rectangle has length **3** and width **8**, its perimeter is less than or equal to **27**.
- B. If the rectangle has length **8** and width **3**, its perimeter is less than or equal to **27**.
- C. If the rectangle has length **3** and width **8**, its perimeter is greater than or equal to **27**.
- D. If the rectangle has length **8** and width **3**, its perimeter is greater than or equal to **27**.

ID: 208626df Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that a rectangle has length ℓ and width w , and the inequality $2\ell + 2w \leq 27$ gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to **27**. To determine the best interpretation of $(\ell, w) = (8, 3)$ in this context, the values can be substituted in the given inequality. Substituting **8** for ℓ and **3** for w in this inequality yields $2(8) + 2(3) \leq 27$, which is equivalent to $16 + 6 \leq 27$, or $22 \leq 27$. Since this inequality is true, if the rectangle has length **8** and width **3**, its perimeter is less than or equal to **27**.

Choice A is incorrect. The interpretation of $(\ell, w) = (8, 3)$ implies that the rectangle has length **8** and width **3**, not length **3** and width **8**.

Choice C is incorrect. The interpretation of $(\ell, w) = (8, 3)$ implies that the rectangle has length **8** and width **3**, not length **3** and width **8**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID f01ef454

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: f01ef454

A geologist estimates that the volume of a slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet. The geologist also estimates that the slab of granite weighs **165** pounds per cubic foot of volume. Which inequality represents this situation, where x represents the estimated total weight, in pounds, of the slab of granite?

- A. $165 - 15.7 < x < 165 - 12.7$
- B. $165 + 12.7 < x < 165 + 15.7$
- C. $165(12.7) < x < 165(15.7)$
- D. $\frac{165}{15.7} < x < \frac{165}{12.7}$

ID: f01ef454 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that the estimated volume of the slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet. It's also given that the estimated weight per cubic foot of volume of that slab is **165** pounds. The estimated total weight of the slab of granite, in pounds, can be calculated by multiplying the estimated volume by the estimated weight per cubic foot. Therefore, if the estimated volume of the slab of granite is **12.7** cubic feet, its estimated total weight is **165(12.7)** pounds, and if the estimated volume of the slab of granite is **15.7** cubic feet, its estimated total weight is **165(15.7)** pounds. Since the estimated volume of the slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet, the estimated total weight x , in pounds, must be greater than **165(12.7)** pounds and less than **165(15.7)** pounds. This situation can be represented by the inequality $165(12.7) < x < 165(15.7)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID e744499e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e744499e

An elementary school teacher is ordering x workbooks and y sets of flash cards for a math class. The teacher must order at least 20 items, but the total cost of the order must not be over \$80. If the workbooks cost \$3 each and the flash cards cost \$4 per set, which of the following systems of inequalities models this situation?

- A.

$x + y \geq 20$

$3x + 4y \leq 80$
- B.

$x + y \geq 20$

$3x + 4y \geq 80$
- C.

$3x + 4y \leq 20$

$x + y \geq 80$
- D.

$x + y \leq 20$

$3x + 4y \geq 80$

ID: e744499e Answer

Correct Answer:
A

Rationale

Choice A is correct. The total number of workbooks and sets of flash cards ordered is represented by $x + y$. Since the teacher must order at least 20 items, it must be true that $x + y \geq 20$. Each workbook costs \$3; therefore, $3x$ represents the cost, in dollars, of x workbooks. Each set of flashcards costs \$4; therefore, $4y$ represents the cost, in dollars, of y sets of flashcards. It follows that the total cost for x workbooks and y sets of flashcards is $3x + 4y$. Since the total cost of the order must not be over \$80, it must also be true that $3x + 4y \leq 80$. Of the choices given, these inequalities are shown only in choice A.

Choice B is incorrect. The second inequality says that the total cost must be greater, not less than or equal to \$80. Choice C incorrectly limits the cost by the minimum number of items and the number of items with the maximum cost. Choice D is incorrect. The first inequality incorrectly says that at most 20 items must be ordered, and the second inequality says that the total cost of the order must be at least, not at most, \$80.

Question Difficulty:
Easy

Question ID 2869fe95

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 2869fe95

For a **3**-week period in a town in Illinois, the lowest recorded temperature was **31** degrees Fahrenheit (**°F**) and the highest recorded temperature was **67°F**. Which inequality is true for any recorded temperature t , in **°F**, in this town for this **3**-week period?

- A. $t \geq 98$
- B. $t \geq 67$
- C. $31 \leq t \leq 67$
- D. $t \leq 31$

ID: 2869fe95 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that for a **3**-week period in a town in Illinois, the lowest recorded temperature was **31°F** and the highest recorded temperature was **67°F**. It follows that the inequality $31 \leq t \leq 67$ is true for any recorded temperature t , in **°F**, in this town for this **3**-week period.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

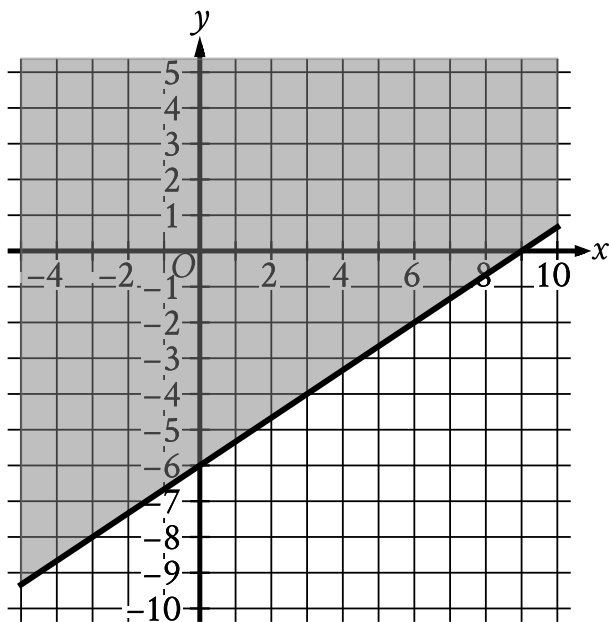
Choice D is incorrect and may result from conceptual errors.

Question Difficulty:
Easy

Question ID c9355dec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: c9355dec



The shaded region shown represents the solutions to which inequality?

- A. $y \geq \frac{2}{3}x - 6$
- B. $y \geq \frac{2}{3}x + 6$
- C. $y \geq \frac{2}{3}x - 9$
- D. $y \geq \frac{2}{3}x + 9$

ID: c9355dec Answer

Correct Answer:
A

Rationale

Choice A is correct. The equation for the line representing the boundary of the shaded region can be written in slope-intercept form $y = mx + b$, where m is the slope and $(0, b)$ is the y-intercept of the line. For the graph shown, the boundary line passes through the points $(0, -6)$ and $(9, 0)$. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, -6)$ and $(9, 0)$ for (x_1, y_1) and (x_2, y_2) , respectively, in this equation yields $m = \frac{0 - (-6)}{9 - 0}$, which is equivalent to $m = \frac{6}{9}$, or $m = \frac{2}{3}$. Since the point $(0, -6)$ represents the y-intercept, it follows that $b = -6$. Substituting $\frac{2}{3}$ for m and -6 for b in the equation $y = mx + b$ yields $y = \frac{2}{3}x - 6$ as the equation of the boundary

line. Since the shaded region represents all the points on and above this boundary line, it follows that the shaded region shown represents the solutions to the inequality $y \geq \frac{2}{3}x - 6$.

Choice B is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, 6)$, not $(0, -6)$.

Choice C is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, -9)$, not $(0, -6)$.

Choice D is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, 9)$, not $(0, -6)$.

Question Difficulty:

Easy

Question ID 84d0d07e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 84d0d07e

A clothing store is having a sale on shirts and pants. During the sale, the cost of each shirt is \$15 and the cost of each pair of pants is \$25. Geoff can spend at most \$120 at the store. If Geoff buys s shirts and p pairs of pants, which of the following must be true?

- A. $15s + 25p \leq 120$
- B. $15s + 25p \geq 120$
- C. $25s + 15p \leq 120$
- D. $25s + 15p \geq 120$

ID: 84d0d07e Answer

Correct Answer:

A

Rationale

Choice A is correct. Since the cost of each shirt is \$15 and Geoff buys s shirts, the expression $15s$ represents the amount Geoff spends on shirts. Since the cost of each pair of pants is \$25 and Geoff buys p pairs of pants, the expression $25p$ represents the amount Geoff spends on pants. Therefore, the sum $15s + 25p$ represents the total amount Geoff spends at the store. Since Geoff can spend at most \$120 at the store, the total amount he spends must be less than or equal to 120. Thus, $15s + 25p \leq 120$.

Choice B is incorrect. This represents the situation in which Geoff spends at least, rather than at most, \$120 at the store. Choice C is incorrect and may result from reversing the cost of a shirt and that of a pair of paints. Choice D is incorrect and may result from both reversing the cost of a shirt and that of a pair of pants and from representing a situation in which Geoff spends at least, rather than at most, \$120 at the store.

Question Difficulty:

Easy

Question ID 89541f9b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 89541f9b

Which of the following ordered pairs (x, y) satisfies the inequality $5x - 3y < 4$?

1. $(1, 1)$

2. $(2, 5)$

3. $(3, 2)$
- A. I only

B. II only

C. I and II only

D. I and III only

ID: 89541f9b Answer

Correct Answer:
C

Rationale
Choice C is correct. Substituting $(1, 1)$ into the inequality gives $5(1) - 3(1) < 4$, or $2 < 4$, which is a true statement. Substituting $(2, 5)$ into the inequality gives $5(2) - 3(5) < 4$, or $-5 < 4$, which is a true statement. Substituting $(3, 2)$ into the inequality gives $5(3) - 3(2) < 4$, or $9 < 4$, which is not a true statement. Therefore, $(1, 1)$ and $(2, 5)$ are the only ordered pairs shown that satisfy the given inequality.

Choice A is incorrect because the ordered pair $(2, 5)$ also satisfies the inequality. Choice B is incorrect because the ordered pair $(1, 1)$ also satisfies the inequality. Choice D is incorrect because the ordered pair $(3, 2)$ does not satisfy the inequality.

Question Difficulty:
Easy

Question ID 915463e0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 915463e0

Normal body temperature for an adult is between 97.8°F and 99°F , inclusive. If Kevin, an adult male, has a body temperature that is considered to be normal, which of the following could be his body temperature?

- A. 96.7°F
- B. 97.6°F
- C. 97.9°F
- D. 99.7°F

ID: 915463e0 Answer

Correct Answer:
C

Rationale

Choice C is correct. Normal body temperature must be greater than or equal to 97.8°F but less than or equal to 99°F . Of the given choices, 97.9°F is the only temperature that fits these restrictions.

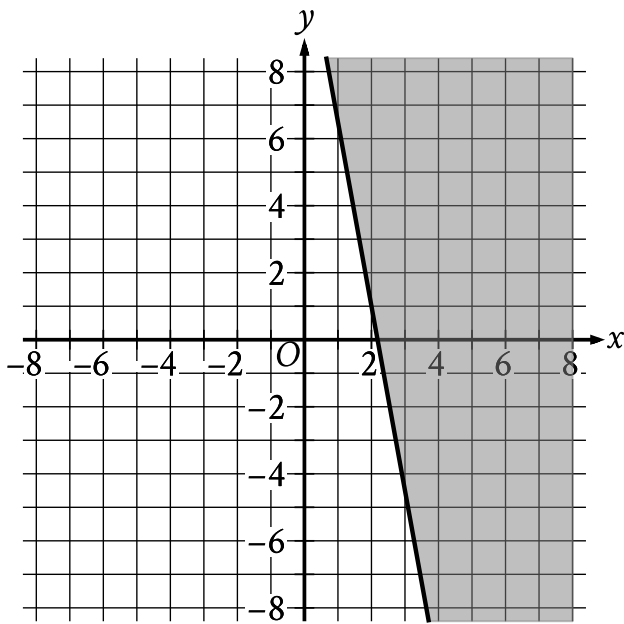
Choices A and B are incorrect. These temperatures are less than 97.8°F , so they don't fit the given restrictions. Choice D is incorrect. This temperature is greater than 99°F , so it doesn't fit the given restrictions.

Question Difficulty:
Easy

Question ID 59a49431

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 59a49431



The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

- A. $(0, -4)$
- B. $(0, 4)$
- C. $(-4, 0)$
- D. $(4, 0)$

ID: 59a49431 Answer

Correct Answer:
D

Rationale

Choice D is correct. Since the shaded region shown represents solutions to an inequality, an ordered pair (x, y) is a solution to the inequality if it's represented by a point in the shaded region. Of the given choices, only $(4, 0)$ is represented by a point in the shaded region. Therefore, $(4, 0)$ is a solution to the inequality.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:

Easy

Question ID e006209c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e006209c

A geologist needs to collect at least **67** samples of lava from a volcano. If the geologist has already collected **63** samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

- A. **130**
- B. **63**
- C. **4**
- D. **0**

ID: e006209c Answer

Correct Answer:

C

Rationale

Choice C is correct. It's given that the geologist has already collected **63** samples from the volcano. Let x represent the number of additional samples the geologist needs to collect. After collecting x additional samples, the geologist will have collected a total of $63 + x$ samples. It's given that the geologist needs to collect at least **67** samples. Therefore, $63 + x \geq 67$. Subtracting **63** from each side of this inequality yields the inequality $x \geq 4$. Thus, the geologist needs to collect a minimum of **4** additional samples.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of samples the geologist has already collected, rather than the minimum number of additional samples the geologist needs to collect.

Choice D is incorrect. If the geologist collects **0** additional samples, the geologist will have collected a total of **63** samples, which is less than **67** samples.

Question Difficulty:

Easy

Question ID 72a5fd28

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 72a5fd28

For a party, **50** dinner rolls are needed. Dinner rolls are sold in packages of **12**. What is the minimum number of packages that should be bought for the party?

ID: 72a5fd28 Answer

Correct Answer:
5

Rationale

The correct answer is **5**. Let ***p*** represent the number of packages of dinner rolls that should be bought for the party. It's given that dinner rolls are sold in packages of **12**. Therefore, **$12p$** represents the number of dinner rolls that should be bought for the party. It's also given that **50** dinner rolls are needed; therefore, **$12p \geq 50$** . Dividing both sides of this inequality by **12** yields **$p \geq \frac{50}{12}$** , or approximately **$p \geq 4.17$** . Since the number of packages of dinner rolls must be a whole number, the minimum number of packages that should be bought for the party is **5**.

Question Difficulty:
Easy

Question ID 86f7483f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 86f7483f

During spring migration, a dragonfly traveled a minimum of **1,510** miles and a maximum of **4,130** miles between stopover locations. Which inequality represents this situation, where *d* is a possible distance, in miles, this dragonfly traveled between stopover locations during spring migration?

- A. $d \leq 1,510$
- B. $1,510 \leq d \leq 4,130$
- C. $d \geq 4,130$
- D. $4,130 \leq d \leq 5,640$

ID: 86f7483f Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that during spring migration, a dragonfly traveled a minimum of **1,510** miles and a maximum of **4,130** miles between stopover locations. It's also given that *d* represents a possible distance, in miles, this dragonfly traveled between stopover locations. It follows that the inequality $1,510 \leq d \leq 4,130$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which a dragonfly traveled a maximum of **1,510** miles between stopover locations.

Choice C is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of **4,130** miles between stopover locations.

Choice D is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of **4,310** miles and a maximum of **5,640** miles between stopover locations.

Question Difficulty:
Easy

Question ID df32b09c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: df32b09c

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G , on Tom’s fourth exam that will result in a mean score on all four exams of at least 90 ?

- A. $90 - (85 + 78 + 98) \leq 4G$
- B. $4G + 85 + 78 + 98 \geq 360$
- C. $\frac{(G + 85 + 78 + 98)}{4} \geq 90$
- D. $\frac{(85 + 78 + 98)}{4} \geq 90 - 4G$

ID: df32b09c Answer

Correct Answer:
C

Rationale

Choice C is correct. The mean of the four scores (G , 85, 78, and 98) can be expressed as $\frac{G + 85 + 78 + 98}{4}$. The inequality that expresses the condition that the mean score is at least 90 can therefore be written as $\frac{G + 85 + 78 + 98}{4} \geq 90$.

Choice A is incorrect. The sum of the scores (G , 85, 78, and 98) isn’t divided by 4 to express the mean. Choice B is incorrect and may be the result of an algebraic error when multiplying both sides of the inequality by 4. Choice D is incorrect because it doesn’t include G in the mean with the other three scores.

Question Difficulty:
Easy

Question ID 68f2cbaf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 68f2cbaf

Ty set a goal to walk at least **24** kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of **4** kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

- A. **4**
- B. **6**
- C. **20**
- D. **24**

ID: 68f2cbaf Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that Ty plans to walk at an average speed of **4** kilometers per hour. The number of kilometers Ty will walk is determined by the expression **4s**, where **s** is the number of hours Ty walks. The given goal of at least **24** kilometers means that the inequality **4s ≥ 24** represents the situation. Dividing both sides of this inequality by **4** gives **s ≥ 6**, which corresponds to a minimum of **6** hours Ty must walk.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 4a090a46

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 4a090a46

Julissa needs at least **100** hours of flight time to get her private pilot certification. If Julissa already has **86** hours of flight time, what is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification?

- A. **14**
- B. **76**
- C. **86**
- D. **186**

ID: 4a090a46 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that Julissa already has **86** hours of flight time. Let x represent the number of additional hours of flight time Julissa needs to get her private pilot certification. After completing x hours of flight time, Julissa will have completed a total of $86 + x$ hours of flight time. It's given that Julissa needs at least **100** hours of flight time to get her private pilot certification. Therefore, $86 + x \geq 100$. Subtracting **86** from both sides of this inequality yields $x \geq 14$. Thus, **14** is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of hours of flight time Julissa already has, rather than the minimum number of additional hours of flight time Julissa needs.

Choice D is incorrect. This is the number of hours of flight time Julissa will have if she completes **100** more hours of flight time, rather than the minimum number of additional hours of flight time Julissa needs.

Question Difficulty:

Easy

Question ID 563407e5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 563407e5

A bakery sells trays of cookies. Each tray contains at least 50 cookies but no more than 60. Which of the following could be the total number of cookies on 4 trays of cookies?

- A. 165
- B. 205
- C. 245
- D. 285

ID: 563407e5 Answer

Correct Answer:
B

Rationale

Choice B is correct. If each tray contains the least number of cookies possible, 50 cookies, then the least number of cookies possible on 4 trays is $50 \times 4 = 200$ cookies. If each tray contains the greatest number of cookies possible, 60 cookies, then the greatest number of cookies possible on 4 trays is $60 \times 4 = 240$ cookies. If the least number of cookies on 4 trays is 200 and the greatest number of cookies is 240, then 205 could be the total number of cookies on these 4 trays of cookies because $200 \leq 205 \leq 240$.

Choices A, C, and D are incorrect. The least number of cookies on 4 trays is 200 cookies, and the greatest number of cookies on 4 trays is 240 cookies. The choices 165, 245, and 285 are each either less than 200 or greater than 240; therefore, they cannot represent the total number of cookies on 4 trays.

Question Difficulty:
Easy

Question ID ee439cff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: ee439cff

On a car trip, Rhett and Jessica each drove for part of the trip, and the total distance they drove was under **220** miles. Rhett drove at an average speed of **35 miles per hour (mph)**, and Jessica drove at an average speed of **40 mph**. Which of the following inequalities represents this situation, where ***r*** is the number of hours Rhett drove and ***j*** is the number of hours Jessica drove?

- A. $35r + 40j > 220$
- B. $35r + 40j < 220$
- C. $40r + 35j > 220$
- D. $40r + 35j < 220$

ID: ee439cff Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that Rhett drove at an average speed of **35** miles per hour and that he drove for ***r*** hours. Multiplying **35** miles per hour by ***r*** hours yields **$35r$** miles, or the distance that Rhett drove. It's also given that Jessica drove at an average speed of **40** miles per hour and that she drove for ***j*** hours. Multiplying **40** miles per hour by ***j*** hours yields **$40j$** miles, or the distance that Jessica drove. The total distance, in miles, that Rhett and Jessica drove can be represented by the expression **$35r + 40j$** . It's given that the total distance they drove was under **220** miles. Therefore, the inequality **$35r + 40j < 220$** represents this situation.

Choice A is incorrect. This inequality represents a situation in which the total distance Rhett and Jessica drove was over, rather than under, **220** miles.

Choice C is incorrect. This inequality represents a situation in which Rhett drove at an average speed of **40**, rather than **35**, miles per hour, Jessica drove at an average speed of **35**, rather than **40**, miles per hour, and the total distance they drove was over, rather than under, **220** miles.

Choice D is incorrect. This inequality represents a situation in which Rhett drove at an average speed of **40**, rather than **35**, miles per hour, and Jessica drove at an average speed of **35**, rather than **40**, miles per hour.

Question Difficulty:
Easy

Question ID c50ede6d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: c50ede6d

The total cost, in dollars, to rent a surfboard consists of a **\$25** service fee and a **\$10** per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of **\$75** to rent the surfboard. Which inequality represents this situation?

- A. $10t \leq 75$
- B. $10 + 25t \leq 75$
- C. $25t \leq 75$
- D. $25 + 10t \leq 75$

ID: c50ede6d Answer

Correct Answer:

D

Rationale

Choice D is correct. The cost of the rental fee depends on the number of hours the surfboard is rented. Multiplying t hours by **10** dollars per hour yields a rental fee of $10t$ dollars. The total cost of the rental consists of the rental fee plus the **25** dollar service fee, which yields a total cost of $25 + 10t$ dollars. Since the person intends to spend a maximum of **75** dollars to rent the surfboard, the total cost must be at most **75** dollars. Therefore, the inequality $25 + 10t \leq 75$ represents this situation.

Choice A is incorrect. This represents a situation where the rental fee, not the total cost, is at most **75** dollars.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 2c121b25

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 2c121b25

Valentina bought two containers of beads. In the first container 30% of the beads are red, and in the second container 70% of the beads are red. Together, the containers have at least 400 red beads. Which inequality shows this relationship, where x is the total number of beads in the first container and y is the total number of beads in the second container?

- A. $0.3x + 0.7y \geq 400$
- B. $0.7x + 0.3y \leq 400$
- C. $\frac{x}{3} + \frac{y}{7} \leq 400$
- D. $30x + 70y \geq 400$

ID: 2c121b25 Answer

Correct Answer:
A

Rationale

Choice A is correct. It is given that x is the total number of beads in the first container and that 30% of those beads are red; therefore, the expression $0.3x$ represents the number of red beads in the first container. It is given that y is the total number of beads in the second container and that 70% of those beads are red; therefore, the expression $0.7y$ represents the number of red beads in the second container. It is also given that, together, the containers have at least 400 red beads, so the inequality that shows this relationship is $0.3x + 0.7y \geq 400$.

Choice B is incorrect because it represents the containers having a total of at most, rather than at least, 400 red beads. Choice C is incorrect and may be the result of misunderstanding how to represent a percentage of beads in each container. Also, the inequality shows the containers having a combined total of at most, rather than at least, 400 red beads. Choice D is incorrect because the percentages were not converted to decimals.

Question Difficulty:
Easy

Question ID 842cec4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 842cec4d

During a portion of a flight, a small airplane's cruising speed varied between **150** miles per hour and **170** miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

- A. $s \leq 20$
- B. $s \leq 150$
- C. $s \leq 170$
- D. $150 \leq s \leq 170$

ID: 842cec4d Answer

Correct Answer:

D

Rationale

Choice D is correct. It's given that during a portion of a flight, a small airplane's cruising speed varied between **150** miles per hour and **170** miles per hour. It's also given that s represents the cruising speed, in miles per hour, during this portion of the flight. It follows that the airplane's cruising speed, in miles per hour, was at least **150**, which means $s \geq 150$, and was at most **170**, which means $s \leq 170$. Therefore, the inequality that best represents this situation is $150 \leq s \leq 170$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID cfe67646

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: cfe67646

The point $(8, 2)$ in the xy -plane is a solution to which of the following systems of inequalities?

- A. $x > 0$
 $y > 0$
- B. $x > 0$
 $y < 0$
- C. $x < 0$
 $y > 0$
- D. $x < 0$
 $y < 0$

ID: cfe67646 Answer

Correct Answer:
A

Rationale

Choice A is correct. The given point, $(8, 2)$, is located in the first quadrant in the xy -plane. The system of inequalities in choice A represents all the points in the first quadrant in the xy -plane. Therefore, $(8, 2)$ is a solution to the system of inequalities in choice A.

Alternate approach: Substituting 8 for x in the first inequality in choice A, $x > 0$, yields $8 > 0$, which is true. Substituting 2 for y in the second inequality in choice A, $y > 0$, yields $2 > 0$, which is true. Since the coordinates of the point $(8, 2)$ make the inequalities $x > 0$ and $y > 0$ true, the point $(8, 2)$ is a solution to the system of inequalities consisting of $x > 0$ and $y > 0$.

Choice B is incorrect. This system of inequalities represents all the points in the fourth quadrant, not the first quadrant, in the xy -plane.

Choice C is incorrect. This system of inequalities represents all the points in the second quadrant, not the first quadrant, in the xy -plane.

Choice D is incorrect. This system of inequalities represents all the points in the third quadrant, not the first quadrant, in the xy -plane.

Question Difficulty:
Easy